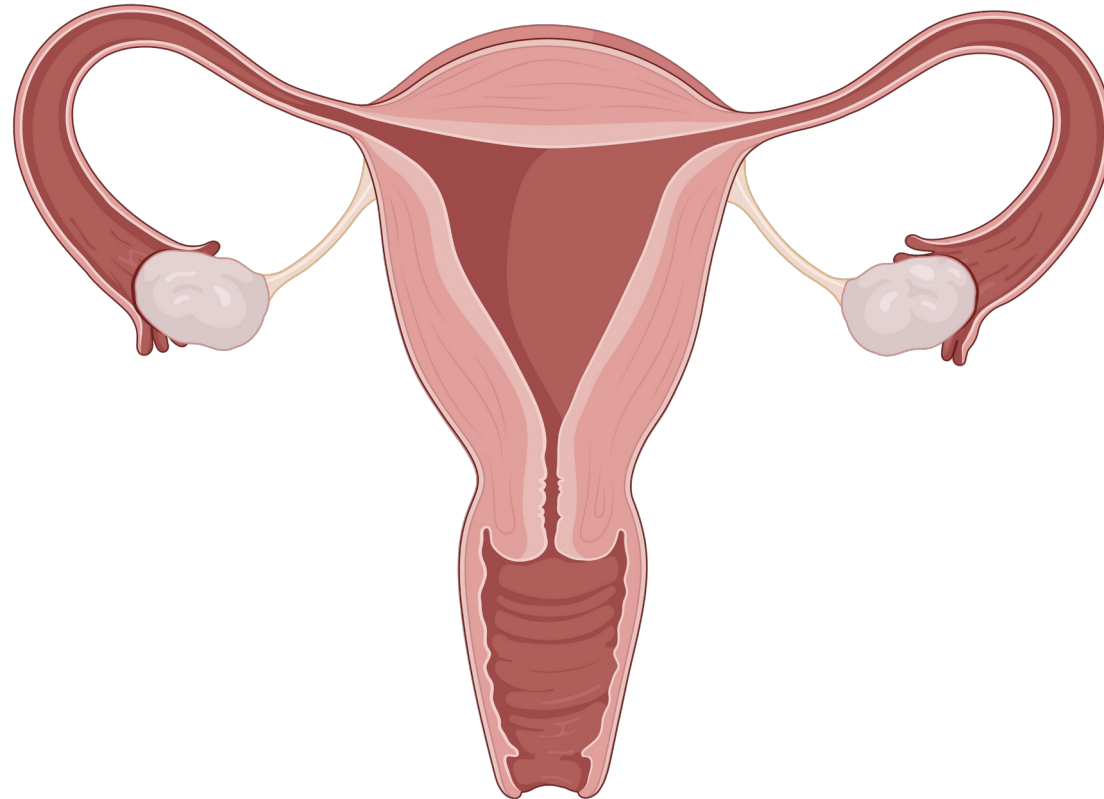


Definitions &
Diagnoses

Hormone
Therapy (MHT)

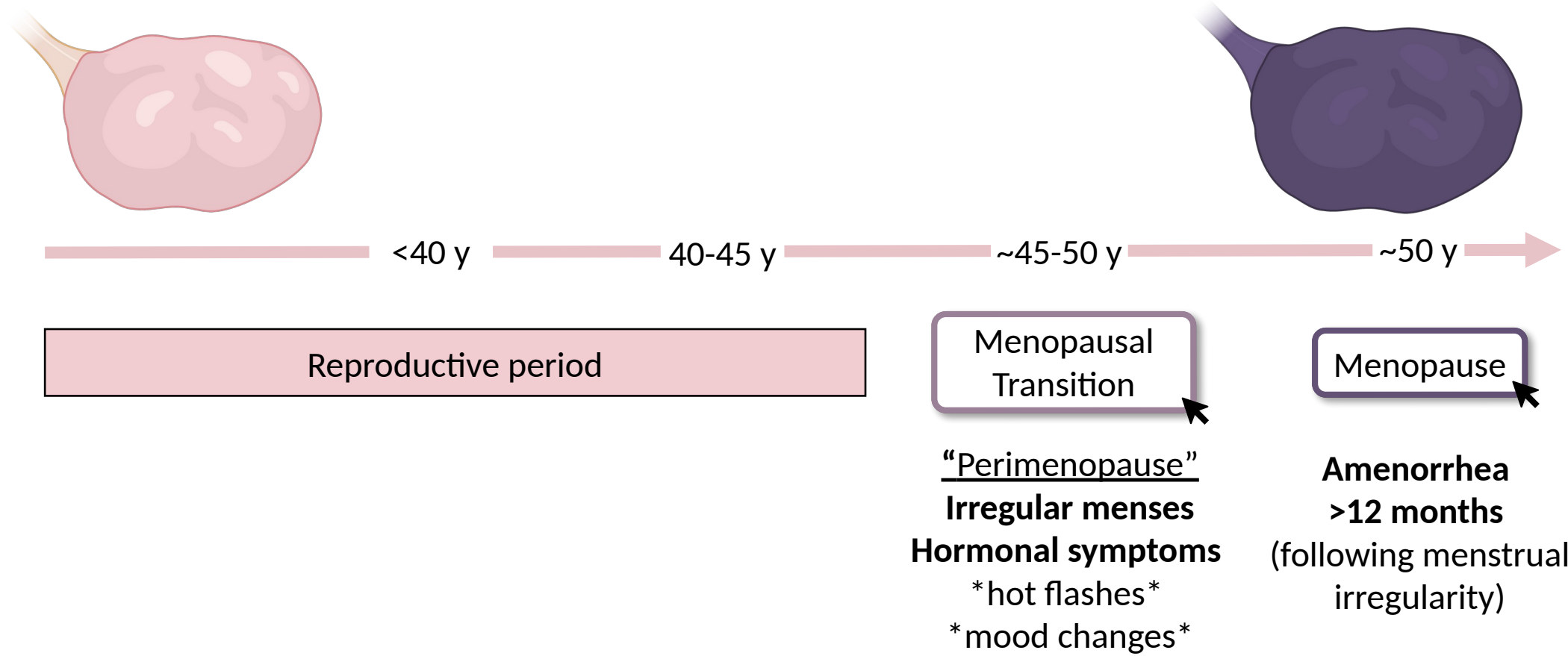
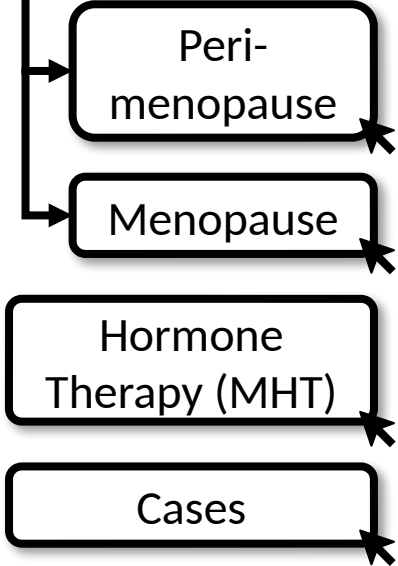
Cases

Perimenopause & Menopause



Clinical Features of Perimenopause and Menopause

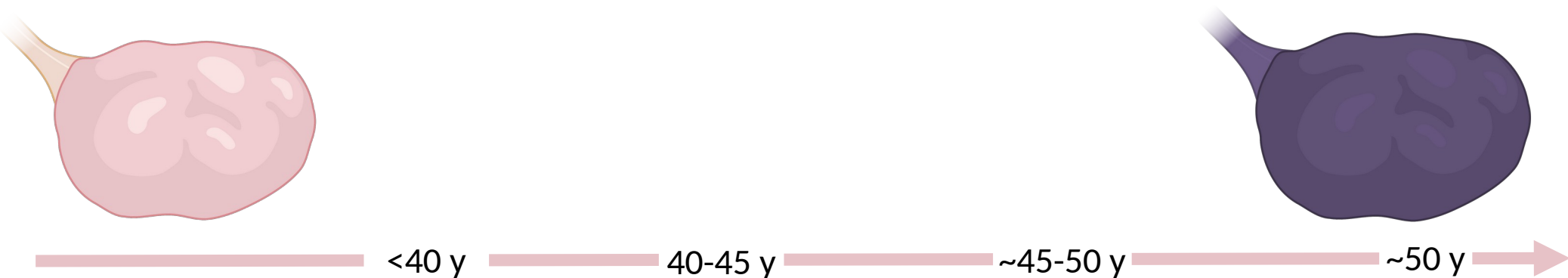
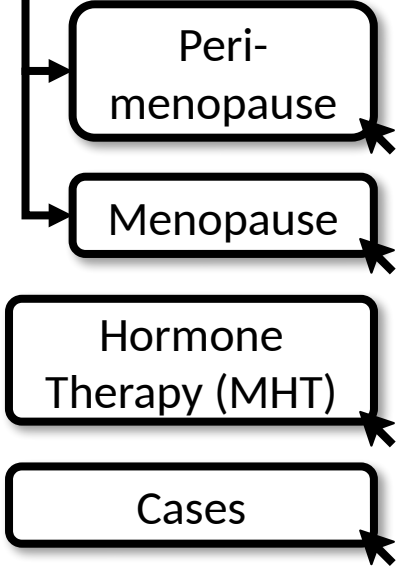
Definitions & Diagnoses



When is Testing Indicated?

Clinical Features of Perimenopause and Menopause

Definitions & Diagnoses



Reproductive period

Primary ovarian insufficiency

Complete work-up for Amenorrhea

Early Menopause

Serum hCG
Prolactin
TSH

Menopausal Transition

“Perimenopause”
Irregular menses
Hormonal symptoms
hot flashes
mood changes

Menopause

Amenorrhea >12 months
(following menstrual irregularity)

When is Testing Indicated?

Clinical Diagnosis

FSH not necessary
(SWAN study. Sowers, 2000)

Menopausal Transition (Perimenopause)

Definitions & Diagnoses

Peri-menopause

Menopause

Hormone Therapy (MHT)

Cases

47 yo woman with hypothyroidism and asthma presents with **hot flashes and difficulty sleeping**. Her menses previously occurred every month. Now they occur **every 2-4 months with heavy flow**.

What additional work-up is indicated?

Rule out malignancy in patients with heavy bleeding!

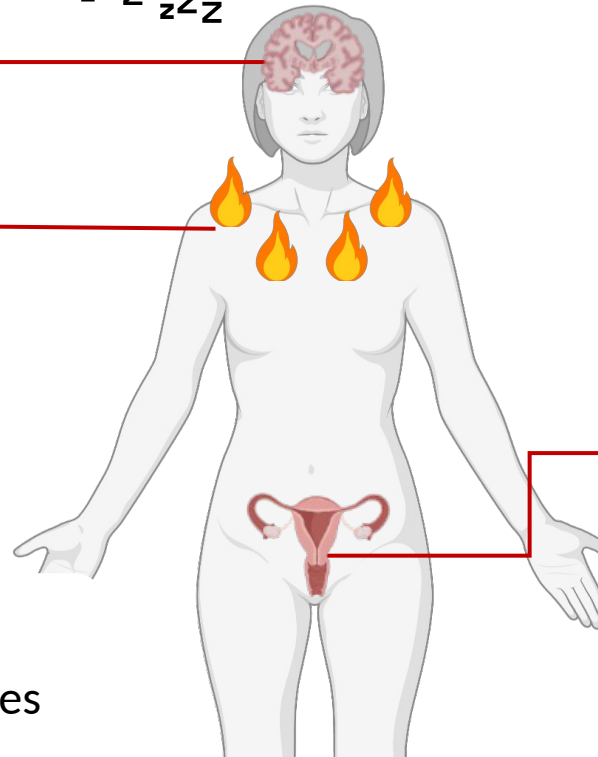
Otherwise, no work-up to confirm diagnosis

How would you manage her symptoms of perimenopause?

Mood Changes & Sleep Disturbance

Hot Flashes

zZz zZz



Menstrual Irregularity

Menopausal hormone therapy
(IUD or combined oral contraceptives)

1. Lifestyle measures

- Avoid hot rooms, hot drinks, EtOH, caffeine and cigarettes.
- Carry a fan & wear layers.
- Weight loss & exercise.

2. SSRI or SNRI, gabapentin

3. Menopausal hormone therapy (MHT) for hot flashes

Definitions & Diagnoses

Peri-menopause

Menopause

Hormone Therapy (MHT)

Cases

Menopause

51yo woman presents with hot flashes and a depressed mood. Her last menses was 14 months ago and was preceded with two years of irregular menses.

What additional work-up is indicated?

None!

How would you manage her symptoms?

Mood Changes,
Sleep Disturbance,
&
Hot Flashes (early)

1. Lifestyle changes
2. MHT (for hot flashes)
3. SSRI or SNRI

Vaginal dryness/ Dyspareunia
UTIs & LUTS

From vaginal atrophy
Vaginal moisturizers, lubricants, topical estrogen

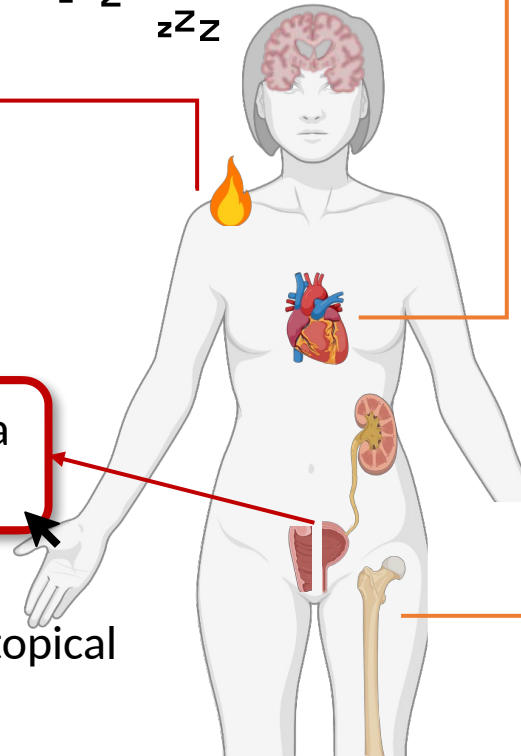
**What complications is she at risk for?
How would you manage them?**

Atherosclerosis

- Primary prevention with:
- Lifestyle modifications
 - $\pm$ statin by ASCVD risk

Osteoporosis

- Primary prevention with:
- Calcium, vitamin D, exercise
 - Smoking cessation
 - $\downarrow$ EtOH



Menopausal Hormone Therapy (MHT)

Definitions & Diagnoses

Hormone Therapy (MHT)

Risk/Benefit

Cases

Who is a candidate?

1. Moderate to severe hot flashes
2. Within 10 y of menopause OR < 60 y of age

Duration of therapy?

5-10 y
Re-evaluate risk/benefit yearly

What is used for MHT?

Estrogen

±

Progesterone

Oral

Transdermal

Vaginal

Fem-ring

Low-dose

	Oral	Transdermal	Fem-ring	Low-dose
VTE/thrombophilia	✗	±	✗	✓
High ASCVD Known CHD/ CVA	✗	✗	✗	✓
Mod ASCVD	±	Preferred	±	✓
Breast cancer	✗	✗	✗	✗
Endometrial cancer	✗	✗	✗	✗
Migraines	±	Preferred	±	✓
Active liver Disease	✗	✗	✗	✓
Vulvovaginal atrophy	✓	✓	✓	Preferred

- For all patients with an intact uterus

✗ = contraindicated

± = discuss risks/benefits

✓ = safe

Definitions &
Diagnoses

Hormone
Therapy (MHT)

Risk/Benefit

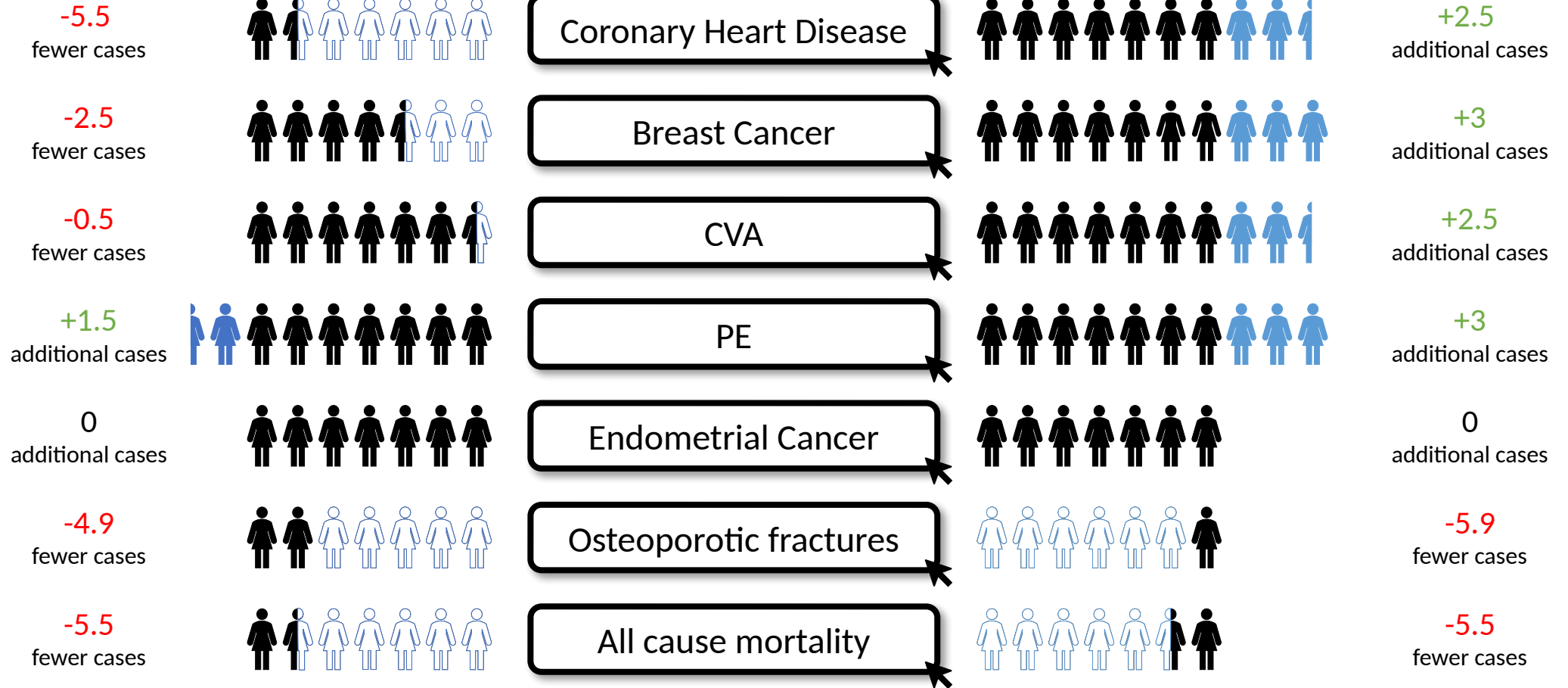
Cases

Menopause Hormonal Therapy (MHT)

For 50-59 yo

Estrogen only

Estrogen + Progesterone



affected per 1000 women
(per 5 y of MHT)

Base incidence of each condition varies and is not represented

Definitions & Diagnoses

Hormone Therapy (MHT)

Cases

1-3

4-6

Case 1

45 yo woman with no PMH presents to clinic with 3 months of amenorrhea, fatigue, and hot flashes.

What are your first steps in evaluation?

- Pregnancy test – women in their 40s are among the highest “at risk group” (second most at risk group) for unplanned pregnancies
- Consider TSH and prolactin (see [AU B talk](#))

You suspect perimenopause. What medications are most effective for hot flashes?

MHT with estrogen and progesterone (if intact uterus).

How will you counsel on the risks of MHT?

Risk of breast cancer, VTE, ACS, stroke.
Plan to stop 5 to 10 years later.

She is sexually active with men. How else will you counsel her?

If she desires to avoid pregnancy, discuss contraception options with patient. (IUD OK to age 55, OCPs to age 51)

Case 2

50 yo woman presents with heavy bleeding. Her periods used to last 3-4 days every month, recently they have been lasting 10-18 days and she has needed to change her sanitary pad every 1-2 hours.

How will you evaluate her menorrhagia?

- Pelvic exam, transvaginal US ± endometrial bx to rule out endometrial cancer.
- Consider FSH, prolactin, TSH to evaluate for perimenopause and thyroid disease.

Workup for other causes are negative, how will you manage her menorrhagia?

Progesterone IUD or OCPs to manage her peri-menopausal menorrhagia

She is interested in using combined OCPs. What are contraindications to use in perimenopause?

- Risk factors for CVD (e.g. smoking, obesity, HTN, DM)
- History of VTE
- Migraines w/ aura
- Breast cancer.

Case 3

49 yo woman with a history of DM, HTN, PAD, tobacco use disorder presents with menopause and asks about hormonal therapy to manage hot flashes.

What lifestyle interventions will you discuss to improve hot flashes?

- Avoid hot rooms, hot drinks, EtOH, caffeine and cigarettes.
- Carry a fan & wear layers.
- Weight loss & exercise.

Would you recommend MHT?

No, MHT is contraindicated in PAD. HTN, DM, and tobacco are all moderate risk when assessed individually, but are high risk in combination.

What other medications may be helpful for her hot flashes?

SNRI/SSRI. Venlafaxine and escitalopram are the most studied. Gabapentin, pregabalin > clonidine are other options.

Definitions & Diagnoses

Hormone Therapy (MHT)

Cases

1-3

4-6

Case 4

48 yo woman with a history of migraine with aura presents with severe hot flashes.

What formulation of MHT would you recommend, if any?

Transdermal estrogen (given migraine with aura) + progesterone.

What risks and benefits are associated with this regimen?

- Slight ↑ risk of breast cancer.
- Slight ↓ risk of fracture, all cause mortality.

Case 5

50 yo woman with a history of TAH (hysterectomy) who presents with hot flashes not responsive to conservative measures. Her ASCVD risk is 7.5%.

What formulation of MHT would you recommend, if any?

Transdermal estrogen (given moderate ASCVD risk). Progesterone is not needed given history of hysterectomy.

What risks and benefits are associated with this regimen?

- No significant risks.
- Slight ↓ risk of CAD, CVA, breast cancer, fracture, and mortality.

Case 6

52 yo woman with a history of breast cancer and frequent UTIs, dyspareunia not responsive to non-hormonal therapies.

What formulation of MHT would you recommend, if any?

Given her history of breast cancer, systemic hormonal therapies are contraindicated. Discuss low dose vaginal estrogen with her oncologist.

What risks and benefits are associated with this regimen?

- Risk of small amount of systemic absorption. Limited data does not show ↑ risk of cancer recurrence, but this should still be discussed with oncology prior to initiation.
- May interfere with aromatase inhibitor efficacy.

Newer Context (2022–2025)

2022 Menopause Society (NAMS) HT Position Statement — reaffirms the deck's rule

Benefits of hormone therapy outweigh risks for most healthy symptomatic women <60 yr OR <10 yr post-menopause; HT remains most effective for VMS and genitourinary syndrome of menopause (GSM), and prevents bone loss/fracture. (NAMS rebranded “The Menopause Society,” 2023.)

New nonhormonal NK-receptor antagonists for moderate–severe hot flashes

Fezolinetant (Vezoh), NK3 antagonist — FDA May 2023 (SKYLIGHT trials). FDA later warned of rare serious liver injury → check LFTs at baseline & periodically.
Elinzanetant (Lynkuet), dual NK1/NK3 antagonist — FDA Oct 2025 (OASIS trials); ~73% VMS reduction, also improves sleep.

2023 Menopause Society Nonhormone Therapy Position Statement

Endorses SSRIs/SNRIs, gabapentin, fezolinetant, CBT, and clinical hypnosis; oxybutynin an option. Vaginal estrogen remains preferred for isolated GSM (minimal systemic absorption).

Take-Home Points

Menopause is primarily a CLINICAL diagnosis based on age and symptoms; lab testing (hCG, TSH, prolactin $\pm$ FSH) is for women <45, or atypical features. FSH is a poor single marker.

Clinical features: menstrual irregularity, vasomotor symptoms (hot flashes), mood/sleep changes, and genitourinary symptoms (dryness, dyspareunia, recurrent UTIs).

Menopausal hormone therapy is the MOST effective treatment for moderate–severe hot flashes; risks are low with good selection (<60 yr or <10 yr post-menopause). Add progesterone whenever the uterus is intact.

Transdermal estrogen has lower VTE/stroke risk than oral (preferred for moderate ASCVD risk, migraine). Vaginal estrogen is preferred for isolated GSM.

Nonhormonal options: SSRI/SNRI (venlafaxine, escitalopram, paroxetine), gabapentin, and now NK-antagonists (fezolinetant, elinzanetant).

Attribution & Changelog

Adapted from TeachIM.org — “Menopause” (Molly Brett MD; sec. ed. Adelaide McClintock MD; exec. ed. Yilin Zhang MD; published Feb 2022).

https://teachim.org/teaching_material/menopause/ · Licensed CC BY-NC 4.0 (non-commercial, attribution required).

Optimized 2026-06-11. No projected-text errors found in the 9-slide board → append-only edit. All click-to-reveal animations & graphics preserved.

Added a newer-context slide (2022 NAMS HT statement; fezolinetant 2023; elinzanetant 2025; 2023 nonhormone statement) — standing teaching unchanged.

Companion files: Menopause-Guide-final.docx (teaching guide); Handout-learner-menopause.pdf. Educational use only — clinician-review before teaching.